Chemical Water Sampling Report: River Towy (LLandeilo - Nant Gurrey) SN 63461 22726

Sampler: Lee Nulty
Date of Sampling: 4th Jun 2026
Time of Sampling: 10:30 AM
River Level (at Manorafon): 1.380 m
Water Temperature: 12.2°C

1. Introduction

This report provides an environmental assessment of the Nant Gurrey, a tributary of the River Towy catchment. The evaluation is based on a localised chemical analysis of the waterbody. The River Towy is a designated Special Area of Conservation (SAC), making the water quality of its tributaries vital for protecting local aquatic ecosystems, fish populations, and macro invertebrates. The purpose of this report is to interpret the provided chemical parameters against UK environmental standards, specifically the Water Framework Directive (WFD)

2. Summary of Chemical Analysis Data

The water samples collected from the Gurrey yielded the following chemical testing results:

- **Total Ammonia:** 1.26 mg/L
- **Nitrate:** 2.6 mg/L
- **Phosphate:** 0.74 mg/L
- **pH:** 8.3

3. Data Interpretation and Environmental Impact

pH (8.3)

The recorded pH of 8.3 indicates that the water is slightly alkaline. This value falls well within the standard regulatory range of 6.0 to 9.0 established for UK river health and is typical for rivers flowing through catchment areas with moderate bicarbonate buffering capacity. At this level, pH does not pose an immediate threat to aquatic life. However, slightly alkaline water can increase the toxicity of un-ionized ammonia, which makes the accompanying ammonia reading a greater concern.

Total Ammonia (1.26 mg/L)

The Total Ammonia level of 1.26 mg/L is significantly elevated and indicates poor water quality. Under WFD standards, unpolluted UK rivers generally maintain annual mean ammonia levels below 0.3 to 0.6 mg/L to achieve "Good" ecological status. A concentration of 1.26 mg/L is toxic to sensitive aquatic species, particularly fish and macro invertebrates. This reading strongly suggests a recent or ongoing input of organic pollution, such as treated or untreated sewage effluent, agricultural slurry, or livestock waste runoff.

Phosphate (0.74 mg/L)

The phosphate concentration of 0.74 mg/L is critically high for a river system. Healthy river systems and target standards set by environmental agencies like Natural Resources Wales typically require phosphate levels to remain below 0.05 to 0.12 mg/L. A reading of 0.74 mg/L represents severe nutrient enrichment (eutrophication). This excess phosphorus stimulates rapid algal growth, which can lead to algal blooms, severe depletion of dissolved oxygen levels, and a subsequent decline in overall river biodiversity. High phosphate levels in this catchment are frequently linked to agricultural fertilisers or domestic wastewater discharges.

Nitrate (2.6 mg/L)

The nitrate concentration of 2.6 mg/L is elevated above baseline natural conditions (which are typically under 1.0 mg/L) but remains well below the legal UK drinking water safety threshold of 50 mg/L. This level is highly characteristic of agricultural catchments where diffuse nitrogen runoff from fertilised fields or soil drainage enters the watercourse. While it does not pose an acute toxicity risk on its own, when combined with high phosphate levels, it accelerates the process of eutrophication.

4. Conclusion

The chemical profile of the River Gurrey highlights significant environmental stress. While the pH is stable and nitrate levels are moderate, both **Total Ammonia** and **Phosphate** levels heavily exceed standard ecological thresholds for UK rivers. The combination of these two elements suggests that the River Gurrey is experiencing a combination of point-source and diffuse pollution —most likely stemming from agricultural activities, failing septic systems, or wastewater discharges.

Because the Gurrey flows directly into the River Towy, these elevated pollutants risk degrading the downstream SAC habitat.